4.2

2) $y'' + 2y' + y = 0$

$y_1 = xe^{-x}$

Let. $y_2 = u y_1$

$= u x e^{-x}$ $\qquad p(x) = 2$

$$= xe^{-x} \int \frac{e^{-\int 2\,dx}}{(xe^{-x})^2} dx$$

$$= xe^{-x} \int \frac{\cancel{e^{-2x}}}{x^2 \cancel{e^{-2x}}} dx$$

$$= xe^{-x} \int \frac{1}{x^2} dx$$

$$= \cancel{x} e^{-x} \left(-\frac{1}{\cancel{x}}\right)$$

$\boxed{y_2 = -e^{-x}}$ $\Rightarrow$ $\boxed{y_2 = e^{-x}}$

constant (-1) ignored

$\frac{(x)^{-2+1}}{-2+1}$

7) $9y'' - 12y' + 4y = 0 \qquad y_1 = e^{2x/3}$

$$y'' - \frac{12}{9}y' + \frac{4}{9}y = 0$$

$$y_2 = u y_1 \Rightarrow u e^{2x/3}$$

$$P(x) = -\frac{12}{9}$$

$$y_2 = e^{2x/3} \int \frac{e^{-\int -\frac{12}{9} dx}}{\left(e^{2x/3}\right)^2} dx$$

$$y_2 = e^{2x/3} \int \frac{e^{\frac{\cancel{12}^{4}}{\cancel{9}_{3}}x}}{e^{4x/3}} dx$$

$$y_2 = e^{2x/3} \int \frac{\cancel{e^{4x/3}}}{\cancel{e^{4x/3}}} dx$$

$$\boxed{y_2 = x e^{2x/3}}$$

14) $x^2 y'' - 3x y' + 5y = 0 \qquad y_1 = x^2 \cos(\ln x)$

$$y'' - \frac{3}{x}y' + \frac{5}{x^2}y = 0 \qquad P(x) = -\frac{3}{x}$$

$$y_2 = u y_1 \Rightarrow x^2 \cos(\ln x) \int \frac{e^{-\int -\frac{3}{x} dx}}{\left(x^2 \cos(\ln x)\right)^2} dx$$

2 $C = P_0$ $\boxed{2 P_0}$ $P_0 e^{kt}$ $\int \tan x$

$\int \frac{\sin x}{\cos x} - \ln\cos x$

$\boxed{y_2 = d}$ d

$$y_2 = x^2 \cos(\ln x) \int \frac{e^{3\ln x}}{x^4 \cos^2(\ln x)} dx$$

$$y_2 = x^2 \cos(\ln x) \int \frac{x^3}{x^4 \cos^2(\ln x)} dx$$

$$= \;'' \;'' \int \frac{1}{x \cos^2(\ln x)} dx$$

$$= \;'' \int \frac{1}{x} \sec^2(\ln x) \, dx$$

$$= \;'' \int \sec^2 t \, dt$$

let $\ln x = t$

$\frac{dt}{dx} = \frac{1}{x}$

$dt = \frac{1}{x} dx$

$$= \;'' \tan(t)$$

$$= x^2 \cancel{\cos(\ln x)} \frac{\sin(\ln x)}{\cancel{\cos(\ln x)}}$$

$$= x^2 \sin(\ln x)$$

16) $(1-x^2) y'' + 2x y' = 0$ $\quad y_1 = 1$

$$y'' + \frac{2x}{1-x^2} y' = 0$$

$y_2 = u y_1 \Rightarrow u \cdot 1$ $\quad P(x) = \frac{2x}{1-x^2}$

$$y_2 = 1 \cdot \int \frac{e^{-\int \frac{2x}{1-x^2} dx}}{(1)^2} dx$$

$$y_2 = \int e^{\ln(1-x^2)} dx \Rightarrow y_2 = \int 1 - x^2 \, dx \Rightarrow \boxed{y_2 = x - \frac{x^3}{3}}$$

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